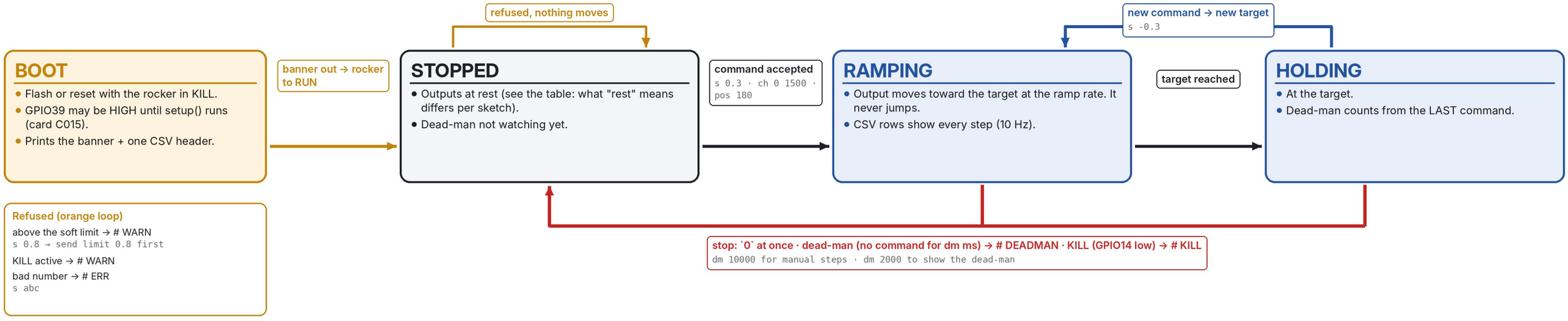
AR-Mini · T3 · Bench sketches: states

firmware/bench · one sketch per module on the DevKit-Lipo · mono = what you type in the serial monitor (115200) · v1 · 10.10.2026

1 · Actuator sketches — bts7960 · pca9685 · st3215 share the same four states



What differs per sketch

	bts7960 · track motor	pca9685 · hobby servo	st3215 · bus servo (arm)
start command	`s <-1..1>`	`ch <n> <µs> · sweep 1`	`pos <deg>`
ramp	0.5 per second (duty)	500 µs per second	60° per second
soft limit → unlock	`|s| ≤ 0.60 → limit 1.0`	`1000–2000 µs → limit 500 2500`	`72°–288° → limit 0 360`
STOPPED means	duty 0 on both pins: the motor coasts	no pulse on all 16 channels: servo limp	halt IN PLACE, torque stays: the arm does not drop (torque 0 = limp)
extra	side L|R stops first, then switches pins	osc <Hz> trims the 50 Hz frame	scan · read · id <n> · resync on the next pos
CSV columns	`t_ms,cmd,pwm_r,pwm_l,is_r_mV,is_l_mV,kill`	`t_ms,ch,us,sweep,kill`	`t_ms,id,cmd_deg,pos_deg,load_permille,temp_C,volt_V,ok`

2 · Sensor sketches — mlx90640 · bno055 · ladder: no states, they only stream

mlx90640

Thermal camera on Wire1. CSV at 4 Hz: min / max / mean °C + hottest pixel; every 2 s an ASCII heat map. raw 1 streams all 768 values.

bno055

IMU on Wire. CSV at 10 Hz: heading, roll, pitch + calibration 0–3 for sys / gyro / acc / mag. cal prints the offsets.

ladder

VBAT divider, leak probe, KILL rocker. CSV at 10 Hz. cal 12.00 with the meter reading; leakth <mV> moves the leak threshold.

3 · The logger

```
python3 scripts/bench_log.py --port /dev/tty.usbmodem* --card C005 --name bts-bench
```

→ data/logs/<date>_C005_bts-bench.csv (never overwrites: adds _2, _3) · what you type is sent to the board AND saved as "# > s 0.3" · Ctrl-C closes the file cleanly · --fake makes 3 s of test data without hardware.